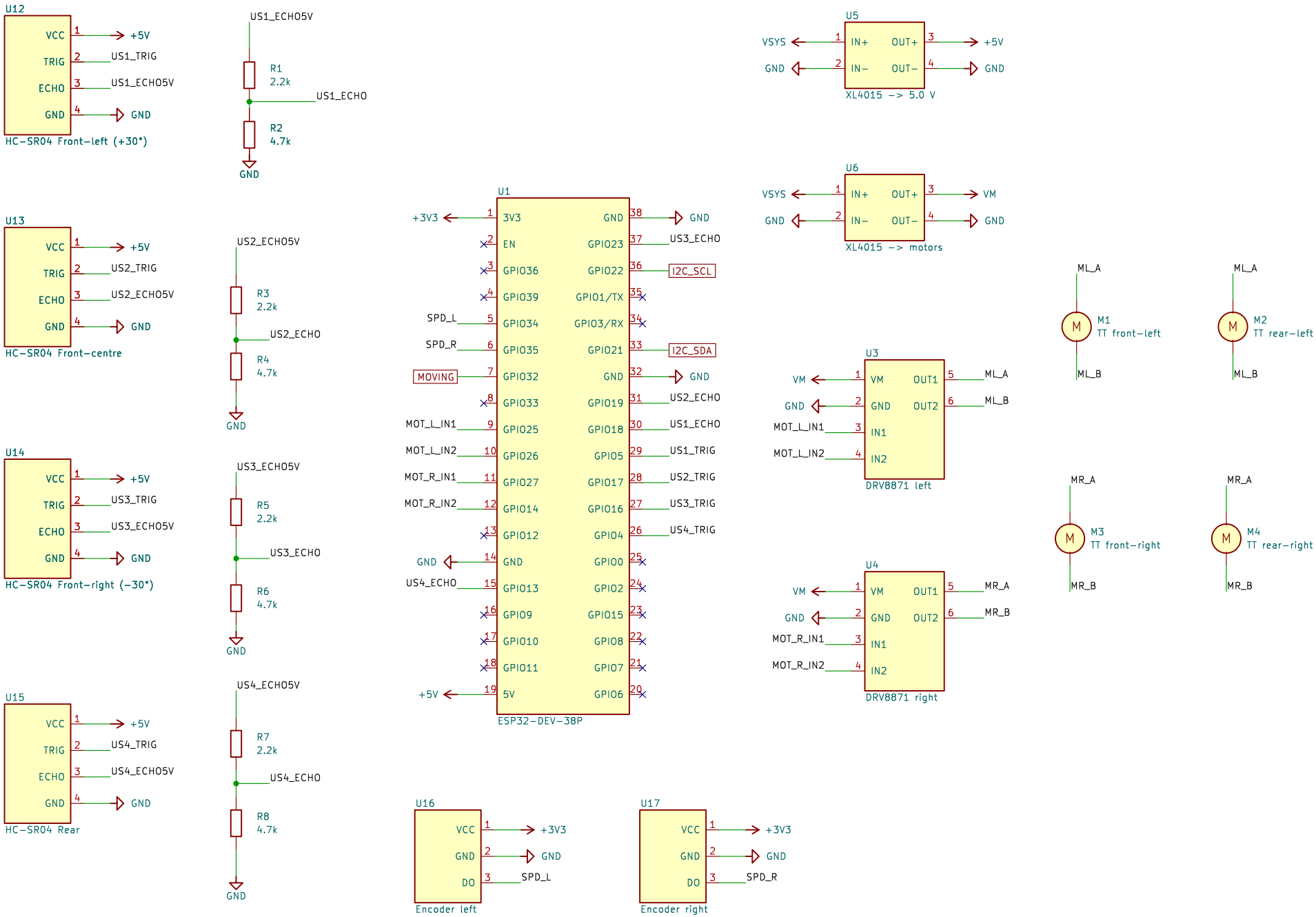

U12
VCC
TRIG
ECHO
GND
+5V
US1_TRIG
US1_ECHO5V
GND
HC-SR04 Front-left (+30°)
US1_ECHO5V
R1
2.2k
US1_ECHO
R2
4.7k
GND
U13
VCC
TRIG
ECHO
GND
+5V
US2_TRIG
US2_ECHO5V
GND
HC-SR04 Front-centre
US2_ECHO5V
R3
2.2k
US2_ECHO
R4
4.7k
GND
U14
VCC
TRIG
ECHO
GND
+5V
US3_TRIG
US3_ECHO5V
GND
HC-SR04 Front-right (-30°)
US3_ECHO5V
R5
2.2k
US3_ECHO
R6
4.7k
GND
U15
VCC
TRIG
ECHO
GND
+5V
US4_TRIG
US4_ECHO5V
GND
HC-SR04 Rear
US4_ECHO5V
R7
2.2k
US4_ECHO
R8
4.7k
GND
U1
3V3
EN
GPIO36
GPIO39
GPIO34
GPIO35
GPIO32
GPIO33
GPIO25
GPIO26
GPIO27
GPIO14
GPIO12
GND
GPIO13
GPIO9
GPIO10
GPIO11
5V
GPIO23
GPIO22
GPIO1/TX
GPIO3/RX
GPIO21
GND
GPIO19
GPIO18
GPIO5
GPIO17
GPIO16
GPIO4
GPIO0
GPIO2
GPIO15
GPIO8
GPIO7
GPIO6
+3V3
SPD_L
SPD_R
MOVING
MOT_L_IN1
MOT_L_IN2
MOT_R_IN1
MOT_R_IN2
GND
US4_ECHO
+5V
GND
US3_ECHO
I2C_SCL
I2C_SDA
GND
US2_ECHO
US1_ECHO
US1_TRIG
US2_TRIG
US3_TRIG
US4_TRIG
ESP32-DEV-38P
U5
VSYS
GND
IN+
IN-
OUT+
OUT-
+5V
GND
XL4015 -> 5.0 V
U6
VSYS
GND
IN+
IN-
OUT+
OUT-
VM
GND
XL4015 -> motors
U3
VM
GND
MOT_L_IN1
MOT_L_IN2
VM
GND
IN1
IN2
OUT1
OUT2
ML_A
ML_B
DRV8871 left
U4
VM
GND
MOT_R_IN1
MOT_R_IN2
VM
GND
IN1
IN2
OUT1
OUT2
MR_A
MR_B
DRV8871 right
ML_A
M
M1
TT front-left
ML_B
ML_A
M
M2
TT rear-left
ML_B
MR_A
M
M3
TT front-right
MR_B
MR_A
M
M4
TT rear-right
MR_B
U16
VCC
GND
DO
+3V3
GND
SPD_L
Encoder left
U17
VCC
GND
DO
+3V3
GND
SPD_R
Encoder right